

10 Biology (Genetics)

ACSSU184: Transmission of heritable characteristics from one generation to the next involves DNA and genes

Week	Australian Curriculum	Achievement	Activities	Assessment
1	Describing the role of DNA as the blueprint for controlling the characteristics of organisms Using models and diagrams to represent the relationship between DNA, genes and chromosomes	DNA the code for life: • What is DNA? / What does it do? • Structure of DNA • How did we find out about DNA?	Variation in our class Modelling DNA	Graphing assessment (3%)
2	Recognising that genetic information passed on to offspring is from both parents by meiosis and fertilisation	DNA Synthesis • How/Why does DNA replicate itself? • How do we relate chromosomes to DNA? Protein Synthesis: • Transcription • Translation Passing DNA onto offspring: • Somatic/gametes • Mitosis/meiosis	'Synthesising' our own proteins. Cell cycle diagrams Amoeba sisters	
3	Representing patterns of inheritance of a simple dominant/recessive characteristic through generations of a family Predicting simple ratios of offspring genotypes and phenotypes in crosses involving dominant/recessive gene pairs or in genes that are sex-linked	How is DNA inherited? • Mendel • Law of segregation & Law of independent assortment Monohybrid crosses: • Practicing drawing monohybrid crosses • Predicting % chance of offspring Sex linkage: • Sex-linked conditions • Communicating sex-linkage Co-Dominance & Incomplete Dominance • What's the difference?	Punnet squares Hypothetical babies Colourblindness Genetics in Harry Potter	
4		Pedigrees: • Reading pedigrees • Keys/symbols used in pedigrees • Drawing pedigrees	Drawing pedigrees	Breeding Beetles Assessment (3%)
5	Describing mutations as changes in DNA or chromosomes and outlining the factors that contribute to causing mutations	Mutations: • Mutations and mutagens • Chromosomal mutations • Case study - cystic fibrosis & Down's Syndrome REVISION	Topic Revision	Topic Test (4%)